specific use case available from third-party vendors is reduced, compared to developing these technologies "in-house."

Further cost-savings are possible by not purchasing co-location services at all major exchanges, and increasing the number of data access functions outsourced. Some market participants may obtain their entire market data feed from a third-party aggregator in the form of a pre-aggregate feed, saving money but surrendering significant ability customize the data feed.[1715] Additionally, some market participants may use the exclusive SIPs instead of proprietary feeds for some use cases. For example, proprietary feeds might only be used for actual order routing decisions, while the exclusive SIPs are used to fill other data needs. Because of the substantial difference in price between exclusive SIP and proprietary feeds, this method may represent a substantial cost savings.[1716] While the benefits of speed and quality of processing may be diminished for those market participants utilizing these more low-cost options, there continue to be trading and order routing strategies for which these approaches are sufficient. However, as acknowledged elsewhere,[1717] execution using these data aggregation methods may experience higher execution costs on average.

1715 See Exegy blog, supra note 1712, describing the affordability of API data feed options and the possibility that customization is reduced relative to less processed options. See also NBIM Letter at 4, stating that broker-dealers who do not perform the aggregation "in-house" will not be "consistently competitive." This commenter also states that "[t]his does not preclude using third-party technology to do the data aggregation, as long as it is done in-house to avoid incremental latency."

1716 For a discussion of the differences in price between exclusive SIPs and proprietary feeds, see supra Section V.B.2(c).

1717 See infra Section V.B.3(e).

One commenter argued that the exclusive SIP feeds could not be used to route orders electronically, stating that "[d]ue to its limited content and higher latency, the usage of SIP data is adequate only for investors that visually consume NMS information (e.g., humans looking at quotes on a screen)."[1718] While the Commission agrees that many users of display feeds use the exclusive SIPs (as discussed in the text below), the Commission believes that there are likely a few non-display users of the exclusive SIP data who route orders based on exclusive SIP feeds as well.[1719]

At the bottom of the continuum are those market participants for which latency sensitivity is not an issue. These include market participants that use human traders who obtain market data through display feeds, and retail investors.[1720] Such market participants frequently outsource the entire data aggregation and dissemination process, including the production of the visual display, to third-party vendors. These market participants also often rely on the exclusive SIP feeds or TOB feeds instead of the DOB feeds. Since latency sensitivity is not an issue, the primary benefit for this type of user of DOB feeds as compared to SIP or TOB feeds is the additional available data. Because of the challenges in obtaining high execution quality using only display feeds and per quote feeds,[1721] many market participants in this last level route their orders to a broker-dealer at a higher level of capability in market data access for execution.

1718 See T. Rowe Price Letter at 1.

1719 For example, one commenter suggested that exclusive SIP feeds play an important role in the activities of some broker-dealers. See Bestex Research Letter at 3.

1720 One commenter stated that the exclusive SIPs are "…the primary feed for retail investors." See Schwab Letter at 2.

1721 See infra Section V.B.3(e) for a discussion of the need for sophisticated use of market data to achieve high quality execution.

Market participants who engage in this type of behavior include investment funds and retail investors. Sometimes broker-dealers working on behalf of clients route orders to a different broker for execution.

One commenter stated "…the Commission fails to consider that proprietary market data is neither necessary nor relevant to the business models and trading or investment strategies of many, if not most, ordinary investors and market participants."[1722] While the Commission acknowledges that this final tier of consumers of market data, which includes most retail investors, might currently rely solely on the SIPs for their own use (and this use might include visual display), the Commission disagrees that proprietary data does not matter to most market participants. The Commission continues to believe that the market participants described as using proprietary data feeds in this section do indeed need those feeds to be competitive with their peers[1723] and that these participants represent a significant segment of the market.[1724] Many market participants, in routing orders to the exchanges, rely on the more sophisticated users of market data to execute orders on their behalf. In other words, while not every individual market participant uses proprietary data feeds, nearly all orders entered into the National Market System, including retail orders, touch a component (typically the order router of the executing broker)

1722 Nasdaq Letter IV at 8.

1723 For additional details on the uses of proprietary data to be competitive, see infra Section V.B.3(e).

1724 While these market participants are less numerous than, for example, retail investors, this does not mean that their role in the market is less significant. For additional insight on this point, see note 1794, which discusses Commission analysis that showed that 91.6% of the message volume on exchanges in a sample week came from just 50 firms. Each of these firms maintained a connection to at least all but one exchange of the 11 exchanges in the sample, which correlates with a relatively high level of sophistication in trading. The fact that the percentage of orders that comes from these firms is so high indicates their significance.

that uses proprietary data in order to reduce execution costs and improve execution quality. The Commission therefore disagrees with this comment that proprietary data is not relevant to the business models of most market participants because the systems by which a broad range of market participants access exchange trading (namely, the network of brokers through which orders are routed) find proprietary data feeds, including their content and speed, relevant to their business models.

(d) Current Costs of Generating SIP Data and Proprietary Data Feeds

As mentioned above,[1725] currently the exclusive SIPs consolidate and disseminate SIP data to market participants. The data fees that exclusive SIPs charge to market participants for obtaining SIP data are set by the Operating Committees of the Equity Data Plans, subject to notice and comment, and Commission approval. A portion of the SIP data revenues is used to pay for the cost of maintaining and administering the exclusive SIP,[1726] and the remaining funds are distributed to the SRO members proportionately to their trading and quoting activity.[1727] In

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1725 See supra Section V.B.1.

1726 Once an exclusive SIP is selected, upgrades to that processor's SIP infrastructure are mandated and funded by the Operating Committee of the relevant Equity Data Plan. This comes out of SIP revenues distributed to the SROs.

1727 The market data revenue allocation formula is summarized at, e.g., UTP Plan, Summary of Market Data Revenue Allocation Formula, available at http://www.utpplan.com/DOC/Revenue_Allocation_Formula.pdf (last accessed Jan. 8, 2020). FINRA rebates a portion of the SIP revenue it receives back to broker-dealer internalizers and ATSs based on the trade volume they report. See FINRA Rule 7610B. One Roundtable commenter estimated that from 2013 to 2017, through the Nasdaq/UTP plan, the FINRA/Nasdaq TRF gave 83 percent of SIP revenue it received to broker-dealers. See Letter from Thomas Wittman, Executive Vice President, Head of Global Trading and Market Services and CEO, Nasdaq Stock Exchange, to Brent J. Fields, Secretary, Commission, at 19 (Oct. 25, 2018).

the case of the UTP SIP, there is an additional FINRA cost for the oversight of the OTC markets that is also taken out of the exclusive SIP's revenues before distributing funds to the plan participants.

Exclusive SIP revenues from data fees totaled more than $430 million in 2017.[1728] There are three broad categories of SIP data fees: access fees, content fees, and distribution/redistribution fees.[1729] An access fee is a flat monthly fee for physical connectivity to SIP data and does not depend on the type of market participant (e.g., market data vendor vs. institutional broker).

There are three categories of content fees that depend on how market participants access SIP data. First, if SIP data is displayed for market participants on computer screens or other devices, the market participant is charged a display fee (a professional or a non-professional subscriber fee depending on the type of market participant). These fees can be per screen displaying the data, per user as part of the multi instance single user (MISU) program, and per application where multiple applications can run on one screen. Second, if SIP data is not displayed on computer screens and instead is directly sent to an automated system such as a trading algorithm or a SOR, then the market participant is charged a non-display fee. Display and non-display fees are monthly fees and entitle the subscriber to an unlimited amount of real-

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1728 See Governance Order, supra note 1128.

1729 See, e.g., CTA Plan, Q2 2020 CTA Quarterly Revenue Disclosure, available at https://www.ctaplan.com/publicdocs/ctaplan/CTA_Quarterly_Revenue_Disclosure_2Q2020.pdf; Nasdaq UTP Plan, Q2 2020 UTP Quarterly Revenue Disclosure, available at https://www.utpplan.com/DOC/UTP_Revenue_Disclosure_Q22020.pdf; see also Letter from Charles M. Jones, Robert W. Lear Professor of Finance and Economics, Columbia Business School, to Brent J. Fields, Secretary, Commission, 15–16 (Oct. 21, 2018) ("Jones Letter").

time market information during the month. In 2018, around 65% to 75% of total SIP revenue was accounted for by professional and non-professional display fees, and around 8% to 13% of revenue was accounted for by non-display fees.[1730] A third type of content fee is the query quote fee, which are fees collected from market participants accessing SIP data on a per quote basis. Under the per-query fee structure, subscribers are required to pay an amount for each request for a packet of real-time market information. Around 4% to 10% of total SIP revenue is accounted for by quote query fees in 2018.[1731] Finally, exclusive SIPs charge distribution/redistribution fees when the market data is delivered to a user other than the initial purchaser.

Based on the exclusive SIPs' public disclosures, as of fourth quarter of 2018 there were approximately 2-3 million non-professional subscription use cases and approximately 0.3 million professional subscription use cases across the UTP and CTA/CQ SIPs. Additionally, there were approximately 300 non-display vendor use cases at each of the exclusive SIPs.[1732] The Nasdaq UTP SIP operating expenses totaled around $7 million in 2017.[1733] The CTA/CQ SIP operating expenses totaled around $8.8 million in 2018.

There is a substantial difference between the fees market participants pay for SIP data and the fees they pay for proprietary DOB data products. For instance, monthly non-display fees

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1730 Id.

1731 Id.

1732 See supra note 1638.

1733 Operating expenses for the Nasdaq UTP Plan represent support costs, paid to the SIP, and are a pre-determined amount agreed upon by the Nasdaq UTP Plan's SRO participants. The Nasdaq UTP SIP costs do not include the costs of the exchanges generating the data they send to the Nasdaq UTP SIP. The UTP Plan also incurs administrative costs and other miscellaneous expenses, which together totaled around $3.6 million.

for data (not including connectivity fees) charged by the CTA/CQ SIP is $2,000 for Network A and $1,000 for Network B,[1734] while monthly non-display fees charged by NYSE for its NYSE Integrated Feed (not including connectivity fees) is $20,000,[1735] which is an order of magnitude larger than the SIP data fee.[1736] Additionally, proprietary data feed fees have increased over the past decade. For instance, SIFMA estimates that between 2010 and 2018 data fees charged by some exchanges went up by three orders of magnitude or more.[1737] In comparison, SIP data fees went up by 5% during the same time period.[1738] Based on Commission staff experience, the

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1734 See CTA Plan, Schedule of Market Data Charges (Jan. 1, 2015), available at https://www.ctaplan.com/publicdocs/ctaplan/notifications/trader-update/Schedule%20Of%20Market%20Data%20Charges%20-%20January%201,%202015.pdf.

1735 See NYSE Proprietary Data Products, Market Data Pricing (Oct. 16, 2020), at 3, available at https://www.nyse.com/publicdocs/nyse/data/NYSE_Market_Data_Pricing.pdf.)

1736 While all of the fees are for non-display purposes, the data content in the CTA/CQ SIP data is different from the NYSE Integrated Feed. The NYSE Integrated Feed is a full DOB feed and provides an order-by-order view of events in the NYSE equities market, whereas the CTA/CQ non-display feed provides consolidated SIP data for Tape A and Tape B securities. See supra Section V.B.2(a) (discussing the difference in content between SIP data and proprietary DOB feeds). See also NYSE, Real-Time Integrated Feed, available at https://www.nyse.com/market-data/real-time/integrated-feed (last accessed Nov. 23, 2020); CTA, Consolidated Tape Association, available at https://www.ctaplan.com/index (last accessed Nov. 23, 2020).

1737 See SIFMA Letter 2018.

1738 SIFMA's study submitted in connection with the Roundtable contained analysis examining the change in fees that some broker-dealers paid for CTA SIP data between 2010 and 2018. The analysis showed that CTA SIP fees for most categories of data increased by an average of 5% between 2010 and 2018. However, the change in the total amount each broker-dealer spent on CTA SIP data varied based on the type of broker-dealer. The analysis found that the average amount of money spent on CTA SIP data by retail broker-dealers declined by 4% between 2010 and 2017, but the average amount spent by institutional broker-dealers increased by 7%. See id. at 21–28.

Commission understands that the number of subscribers to proprietary market data is relatively small.[1739] The Commission understands that the number of subscribers of proprietary market data and proprietary market data revenues vary across exchanges and that some exchanges obtain a larger percentage than other exchanges of their total market data revenue from proprietary data products (as opposed to revenue from SIP data products). For example, the Commission estimates that in 2018, NYSE collected approximately 5% of its net revenues from selling proprietary market data products. On the other hand, according to the Commission's estimates, Cboe BYX collected approximately 9% of its revenues from selling proprietary market data products.[1740]

As mentioned above,[1741] market participants who purchase proprietary data feeds from multiple SROs may choose to self-aggregate multiple data feeds, or, alternatively, they can purchase already consolidated data from market data aggregators. The exchanges charge a data

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1739 See Proposing Release, 85 FR at n.140.

1740 See infra Section V.B.2(e). The Commission estimates are based on NYSE and Cboe BYX's Form 1 filings and UTP and CTA/CQ revenue metrics. NYSE's Form 1 filings disclose $968 million as its net revenues in 2018. NYSE's revenues from the SIP redistribution is approximately $47 million. Note 2 to the exchange's financial statements states that NYSE collects market data revenues from the exclusive SIPs and "to a lesser extent for (sic) New York Stock Exchange proprietary data products," indicating that the approximately $47 million in revenues from SIP data could be a benchmark for their proprietary market data revenues. NYSE Form 1, available at https://www.sec.gov/Archives/edgar/vprr/1900/19003689.pdf (last accessed Jan. 29, 2020). Similarly, Cboe BYX Form 1 filings report $58 million in net revenues. Of this $58 million, $26 million were market data revenue—approximately $21 million from SIP data revenues and $5 million from proprietary market data revenues. Cboe BYX Form 1, available at https://www.sec.gov/Archives/edgar/vprr/1900/19003669.pdf (last accessed Jan. 29, 2020).

1741 See supra Section V.B.2(b).

fee to any market participant that purchases exchanges' data from market data aggregators.[1742] Therefore, these fees are effectively a part of the total price that a market participant must pay when purchasing data from a market data aggregator. In some cases, these fees may be so high that some market participants cannot afford to self-aggregate proprietary feeds from all exchanges or purchase market data aggregator products.[1743] The Commission believes that more active market makers and some sophisticated broker-dealers including a number of HFT firms and some of the larger banks with proprietary data feed trading desks either self-aggregate or purchase aggregation services or products from third-party vendors.

The Commission understands that the data fees the exchanges charge to market participants that purchase the exchanges' data from market data aggregators may account for a significant portion of the total price market participants pay for the market data aggregators' data products. However, the Commission does not have information on the pricing of market data aggregators' data and cannot break down market data product prices between the direct data fees charged by the exchanges and the fees charged by market data aggregators for their services. The Commission stated this lack of information in the Proposing Release and did not receive the information in the comment letters.

The exchanges also charge fees for various connectivity services they offer (e.g., co-location, fiber connectivity, and wireless connectivity). Connectivity services permit a customer to access an exchange's proprietary market data and/or its trading and execution systems as well

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1742 Some exchanges charge redistribution fees or their equivalents to market data aggregators and separately, one or more data fees (based on different use cases such as professional or non-professional, display or non-display) to market participants who purchase the exchanges' data from market data aggregators. See Virtu Letter I at 16–79 (Exhibit "A," lists of data and connectivity fees by several exchanges).

1743 See, e.g., Roundtable Day One Transcript at 128–29 (Mark Skalabrin, Redline Trading Solutions).

as SIP data. The purchase and use of certain connectivity services is necessary to directly access an exchange's market data and to directly participate in that market, at least for those market participants that represent the vast majority of trading activity on exchanges. Additionally, these connectivity services may be needed in order to take advantage of the reduced latencies offered by the proprietary data feeds, including when market participants prefer the contents of SIP data consolidated from the proprietary data feeds—rather than delivered by an exclusive SIP—to avoid additional latencies.

Connectivity fees can be substantial. For instance, the annual fiber connectivity fees per port at the exchanges' primary data centers are $90,000 at Cboe, $120,000 at Nasdaq, and $168,000 at NYSE.[1744] Co-location services may have two components: an initial fee and an ongoing monthly fee based on the kilowatt (kW) usage. For example, at NYSE, an initial fee for a dedicated high-density cabinet that consumes 9kW per month is $5,000, and an ongoing monthly fee per kW is $1,050.[1745] At Nasdaq, an initial fee is $3,500, and an ongoing monthly fee is $4,500.[1746] Thus, for a year of co-location in a dedicated cabinet with 9kW power, these fees add up to over $118,000 for NYSE and over $57,000 for Nasdaq.

1744 See Letter from Brad Katsuyama, CEO, Investors Exchange LLC, to Brent J. Fields, Secretary, Commission, at Table 7 (Jan. 29, 2019) ("Katsuyama Letter II") (10Gb fiber connectivity).

1745 See NYSE price list 2020, supra note 1434.

1746 See Nasdaq, Price List - Trading Connectivity, available at https://www.nasdaqtrader.com/Trader.aspx?id=PriceListTrading2 (last accessed Dec. 19, 2019).